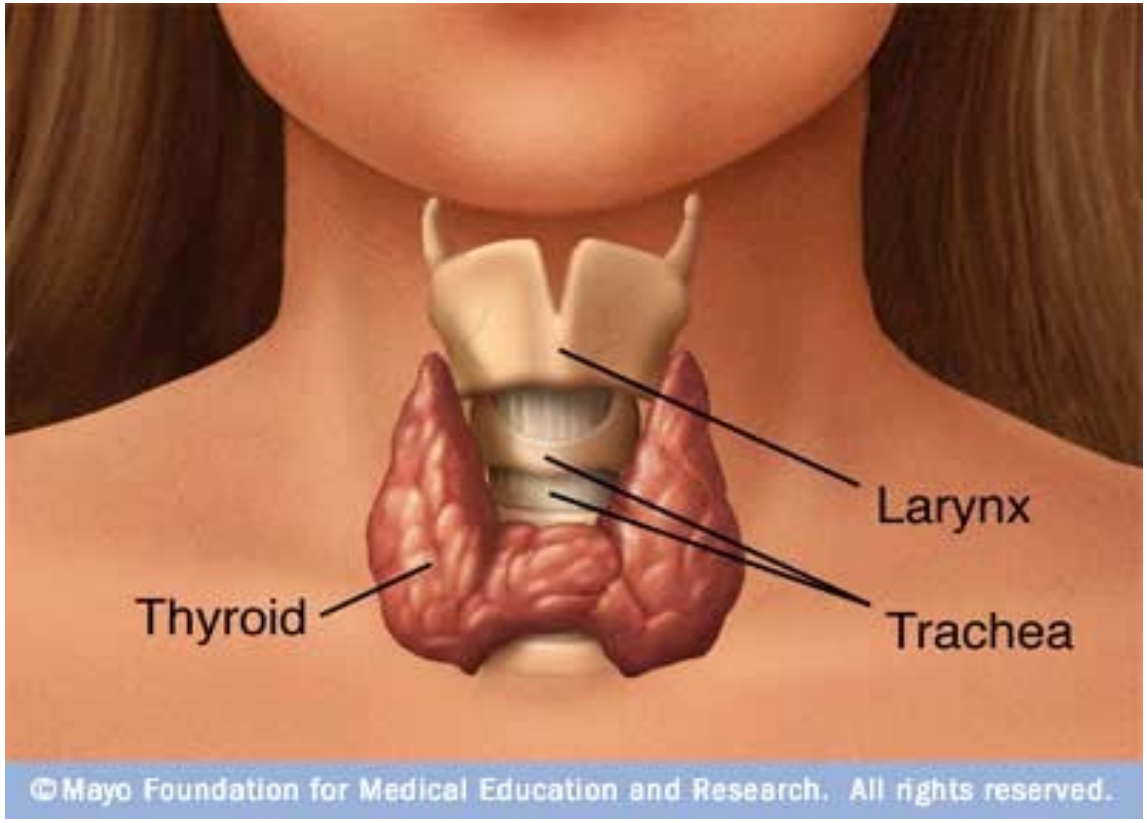
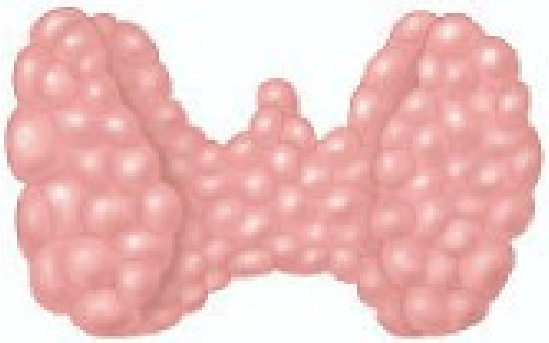
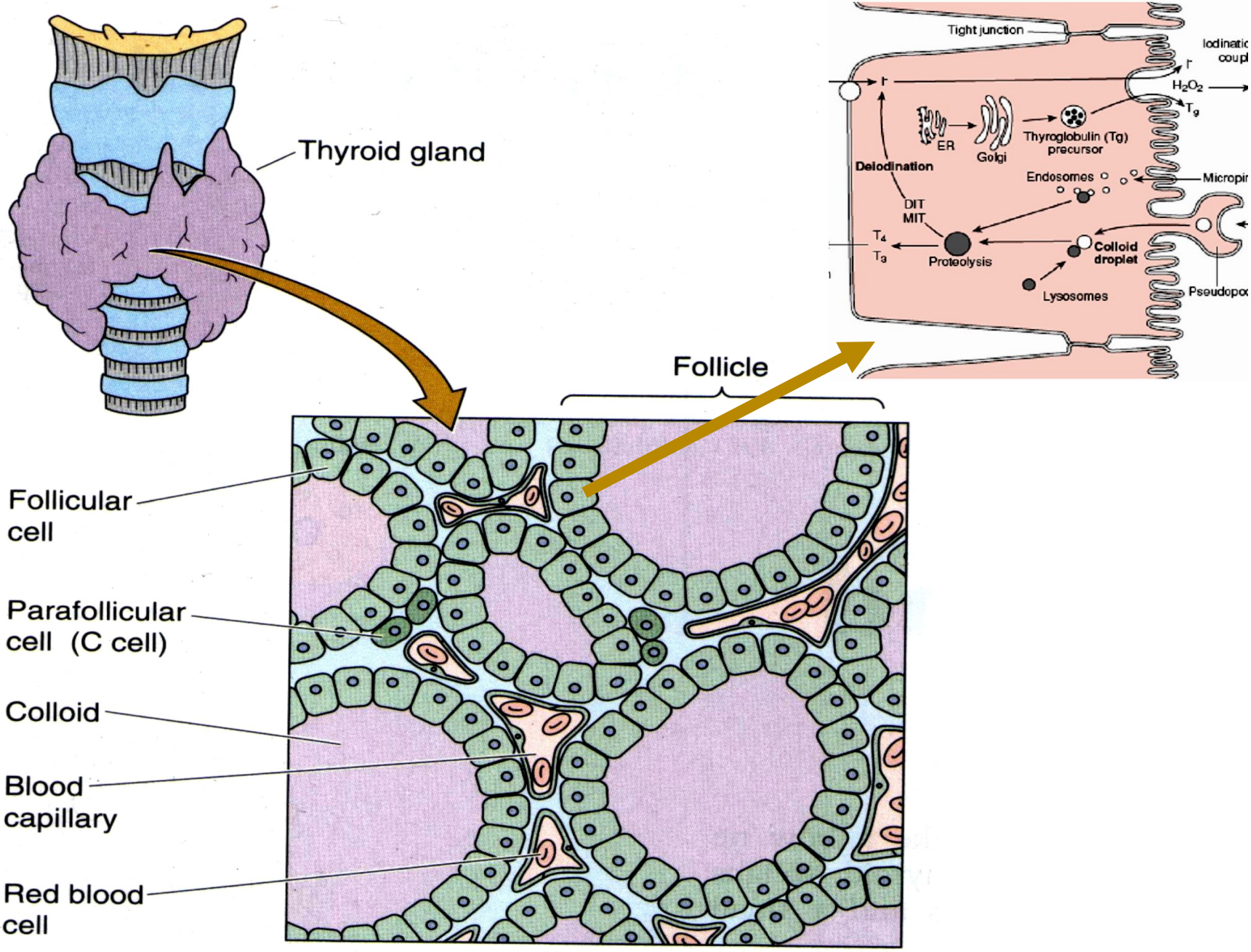
THYROID HORMONE

Gross and Microscopic anatomy





Thyroid


Thyroid gland
Follicle
Follicular cell
Parafollicular cell (C cell)
Colloid
Blood capillary
Red blood cell
Tight junction
Iodinatic
coupl
I⁻
H_2O_2
T_g
ER
Golgi
Thyroglobulin (Tg) precursor
Deiodination
Endosomes
Micropir
DIT
MIT
T_4
T_3
Proteolysis
Colloid droplet
Lysosomes
Pseudopo

Thyroid Hormones

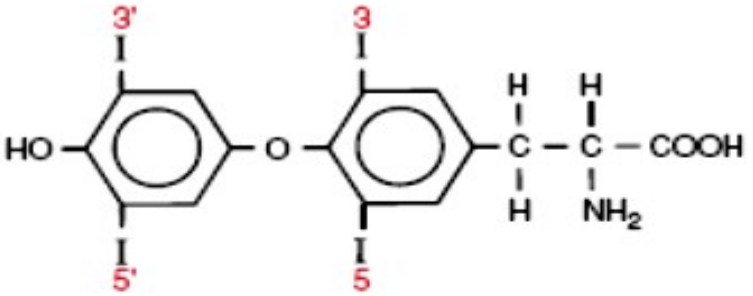
- **The thyroid gland forms & releases:**

1- Triiodothyronine (T3) & tetraiodothyronie (T4) or thyroxine: Secreted by follicular cells.

2- Calcitonin: secreted by parafollicular (C) cells.

Thyroxine (T_4) & triiodothyronine (T_3)

- They are iodinated derivatives of tyrosine amino acid.

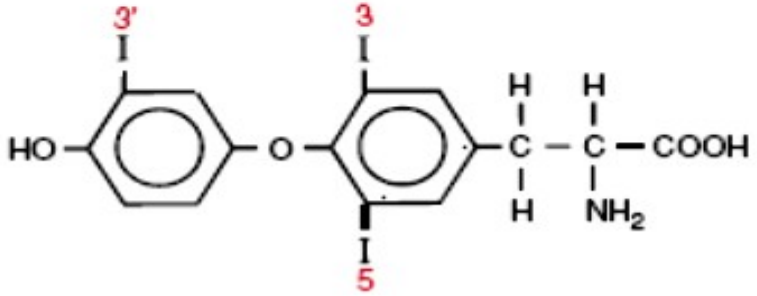


Thyroxine (T_4)

- Thyroid gland secretes:

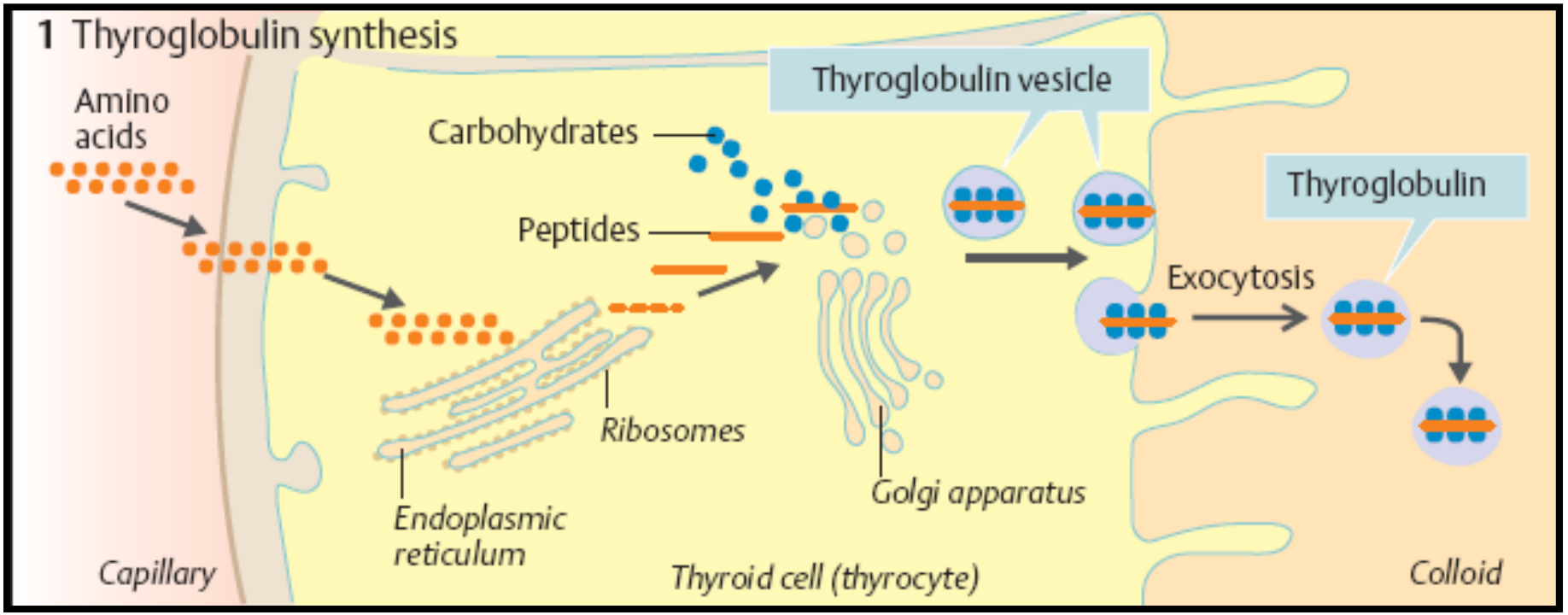
- T_4 (forms 90% of the output).

- T_3 (the remaining 10%)

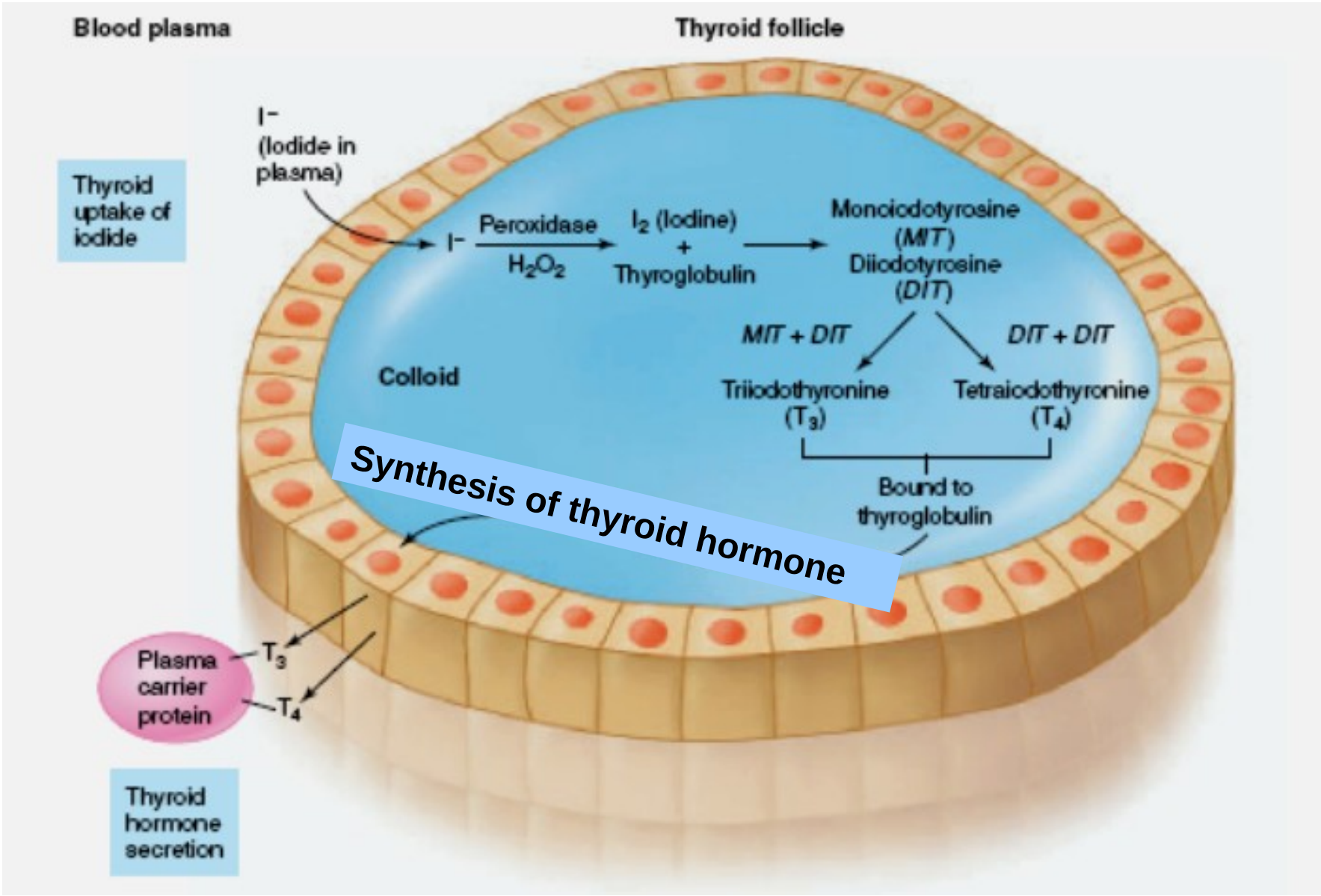


Triiodothyronine (T_3)

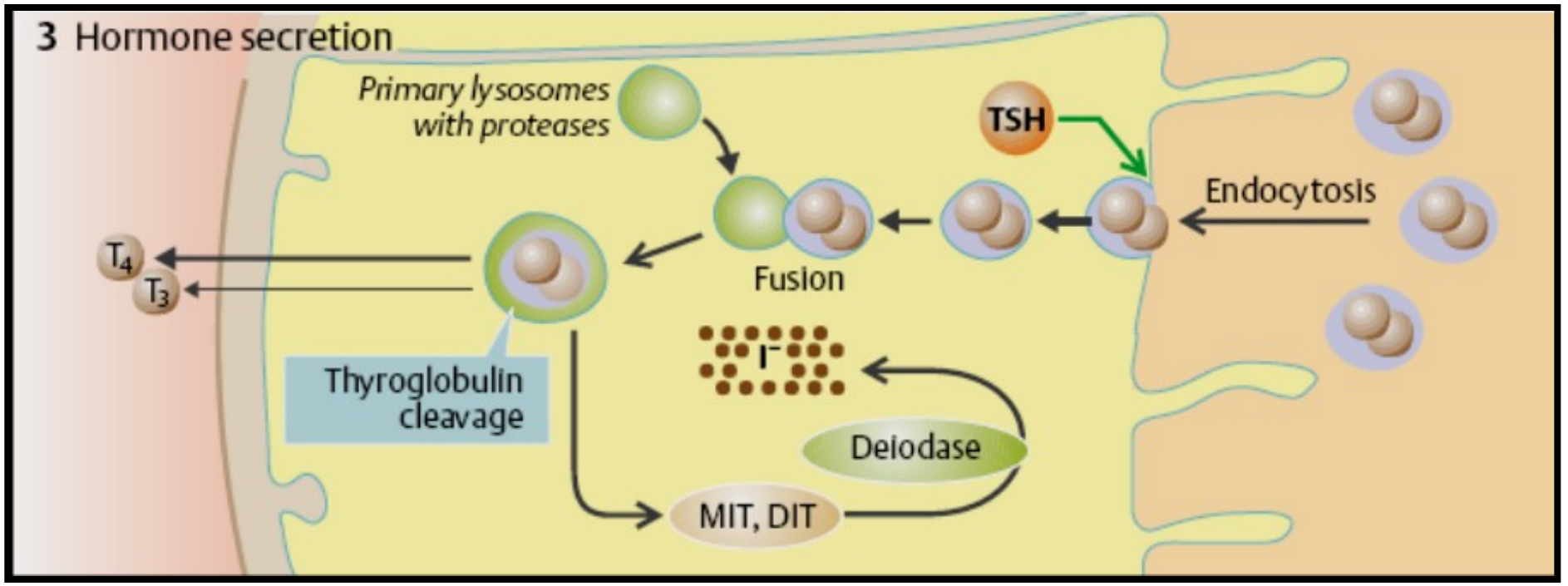
N.B.: Most of serum T_3 comes form peripheral conversion of T_4 in the tissues

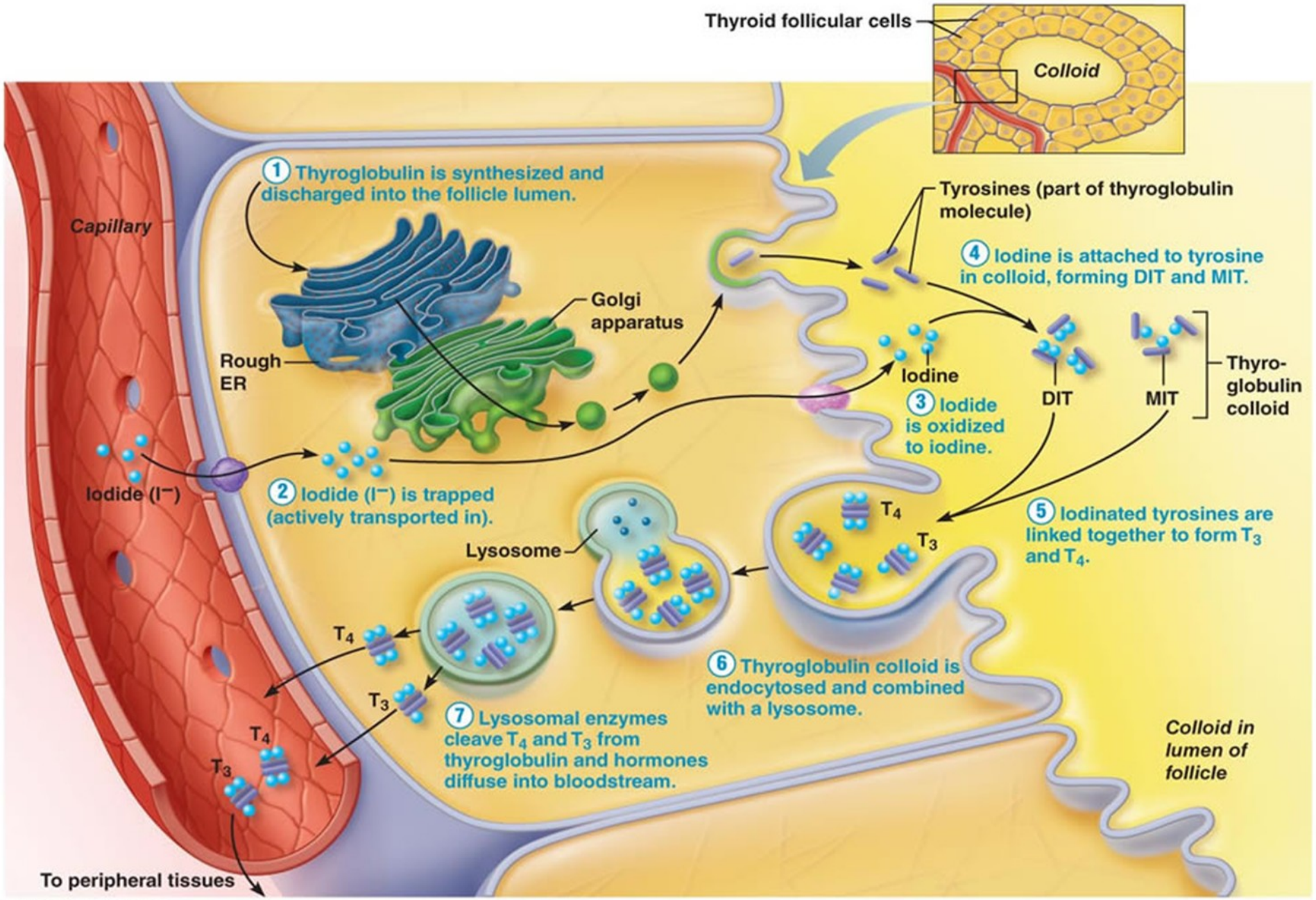


1. **Active uptake of inorganic iodide (I^-).**
2. **Oxidation of I^- to organic iodine I_2.**
3. **Iodination of tyrosine to produce mono & diiodotyrosine.**
4. **Oxidative coupling to form T3 & T4.**


Blood plasma
Thyroid follicle
I⁻
(Iodide in plasma)
Thyroid uptake of iodide
I⁻
Peroxidase
H_2O_2
I_2 (Iodine)
+
Thyroglobulin
Monoiodotyrosine (MIT)
Diiodotyrosine (DIT)
MIT + DIT
DIT + DIT
Colloid
Triiodothyronine (T_3)
Tetraiodothyronine (T_4)
Bound to thyroglobulin
Synthesis of thyroid hormone
Plasma carrier protein
T_3
T_4
Thyroid hormone secretion

Storage & Release





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Transport

Bound	Free	Plasma level	
99.8%	**0.2%**	**0.15 mcg%**	**T3**
99.98%	**0.02%**	**8 mcg%**	**T4**

- ***The transport proteins are:***

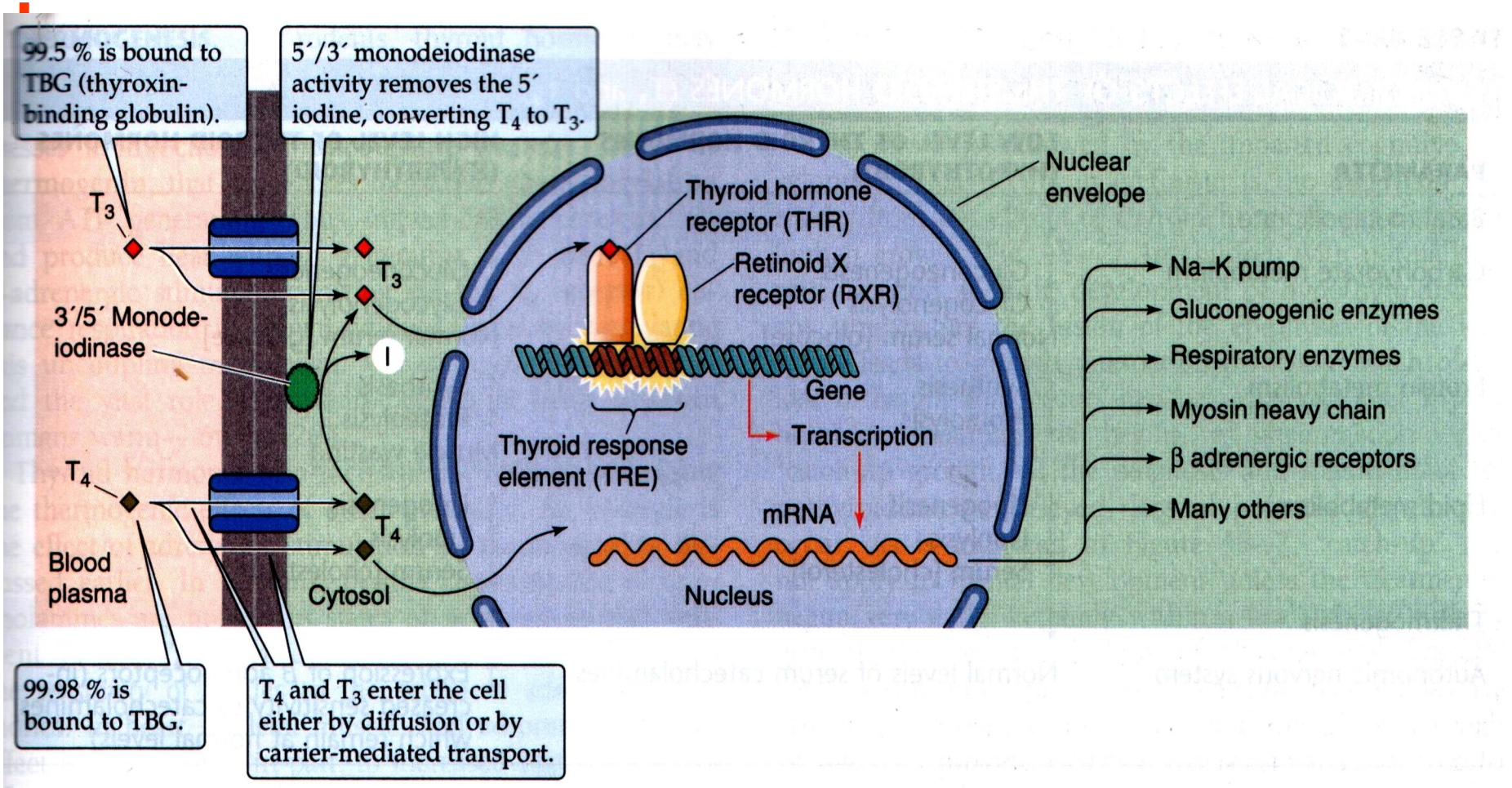
a) Thyroid binding globulin (TBG) (α-globulin) (50%)

b) Thyroxine binding prealbumin (TBPA). (40%)

c) Plasma albumin. (10%).

Functions

1- Intracellular actions of thyroid



1) ↑ size & number of mitochondria -----> ↑ rate of respiration.

2) ↑ activity & amount of membrane Na-K ATPase

3) Stimulate amino acid transport across the cell membrane -----> facilitate protein synthesis.

4) Activates proteolytic & lysosomal enzymes especially in muscles

Effect on teeth

- The common oral findings in hypothyroidism include the characteristic macroglossia, delayed eruption, poor periodontal health, altered tooth morphology and delayed wound .healing

- The oral manifestations of thyrotoxicosis, includes increased susceptibility to caries, periodontal disease, enlargement of extraglandular thyroid tissue (mainly in the lateral posterior tongue), maxillary or mandibular osteoporosis, accelerated dental eruption, and burning mouth syndrome.
- **Burning mouth syndrome**, a condition that causes a burning pain in the mouth, and Sjogren's syndrome, a condition that causes dry mouth, are more common in people with thyroid disease.

2- effect on growth:

a) Physical growth:

- Has a permissive effect on GH & potentiate the effect of somatomedins.

- Eruption & development of teeth. - Closure of fontannels.

b) Mental growth:

- Growth, development & function of CNS during fetal life & 1st few years after birth.

- Myelination of nerve fibers & development of synapses.

C) Sexual growth:

- Essential for normal menestrual cycle & spermatogenesis (fertility).

- Stimulates milk secretion during lactation.

3- Metabolic action:

a) General metabolism:

- 1 mg of thyroxine increase heat production by 1000 calories.

- This is due to increased O_2 consumption & metabolic rate.

- This is called **"calorigenic action"** of thyroid hormone.

- this occurs in all body tissues except: adult brain, retina, gonads, lymphoid tissues & lung.

- In anterior pituitary, thyroid ↓ metabolism due to negative feedback.

b) Protein Metabolism:

- Normal levels of thyroid hormones stimulate protein synthesis & growth (anabolic).

- Excess thyroid hormone causes protein catabolism.

c) Carbohydrate Metabolism:

In physiologic amounts:

a)Enhance action of insulin.

b)Stimulate glycogenesis.

c)Stimulate glucose utilization.

However, in pharmacological doses it will:

- ↑ intestinal absorption of glucose.

- hepatic glycogenolysis b y catecholamines.

- Gluconeogenesis.

d) Lipid Metabolism:

- It stimulates all aspects of lipid metabolism including: synthesis, mobilization & utilization.

It decreases the level of cholesterol, triglycerides and phopholipids in the blood But increases the level of free fatty acid & glycerol.

e) Vitamin & drugs Metabolism:

- hepatic conversion of carotene into vitamin A.

-↑ utilization & clearance of all vitamines.

- ↑clearance of a number of drugs including digitalis.

4- Effect on body systems:

a) Cardiovascular system:

i. Increased heart rate: due to

- stimulation of metabolism of SAN.

- Increased sensitivity of SAN to circulating catecholamines.

- Increased body temperature -------> Peripheral V.D. ------> Increased Venous return (Bainbridge reflex) & hypotension (Mary's law).

ii. Increased strength of myocardial contractility: due to?

iii. Increased cardiac output.

iv. Arterial blood pressure:

- Increased systolic ABP. due to????
- Decreased diastolic ABP. due to????
- Increased pulse pressure.

b) Respiratory system:

Increased rate & depth of respiration due to???????

c) Gastrointestinal tract:

- Stimulate secretion of digestive juices (including gastric HCl). - increased appetite.

- Stimulate intestinal motility & absorption.

d) Hemopoietic system:

i. stimulate erythropoiesis through:

- increased metabolism of bone marrow.

- increased secretion of erythropoitin hormone.

- increased absorption of vitamin B12 from the ileum.

ii. Shift of O_2 dissociation curve to the right due to increased 2,3-DPG formation in RBCs.

e- Endocrine System:

i. negative feedback inhibition on the pituitary TSH.

ii. Synergistic action on catecholamines:

- it increased number of catecholamines receptors & sensitivity to catecholamines.

- it appears that CVS & CNS manifestations of thyroid are due to this synergism.

iii. Permissive effect on glucagon, cortisol & GH.

- Increased sex hormone binding globulin level -----> decreased clearance of sex hormones.

- Increased cortisol clearance in the liver ------> increased ACTH.

f- Skeletal muscle:

- Essential for normal function of the skeletal muscle.

Control of Thyroid Functions

1- Thyroid Stimulating Hormone (TSH)

- a glycoprotein secreted by thyrotrop basophil.

- Actions:

a) stimulate all steps of thyroid hormone secretion & release.

b) increase the size & vascularity of thyroid gland. Follicular cells becomes columnar & active.

N.B.: Prolonged stimulation of the gland by TSH ------> ↑ size (hypertrophy) & number (hyperplasia) of follicular cells -------> enlargement of thyroid gland ***(goiter).***

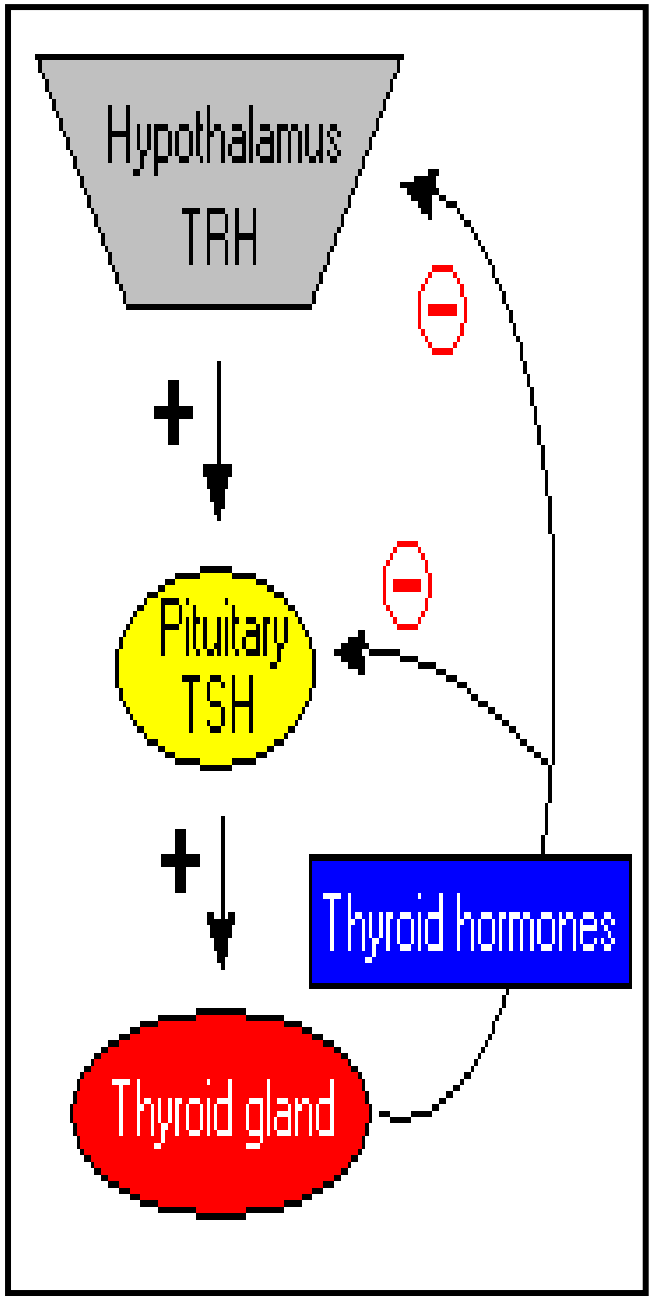
Control of TSH secretion:

1- Hypothalamic TRH:

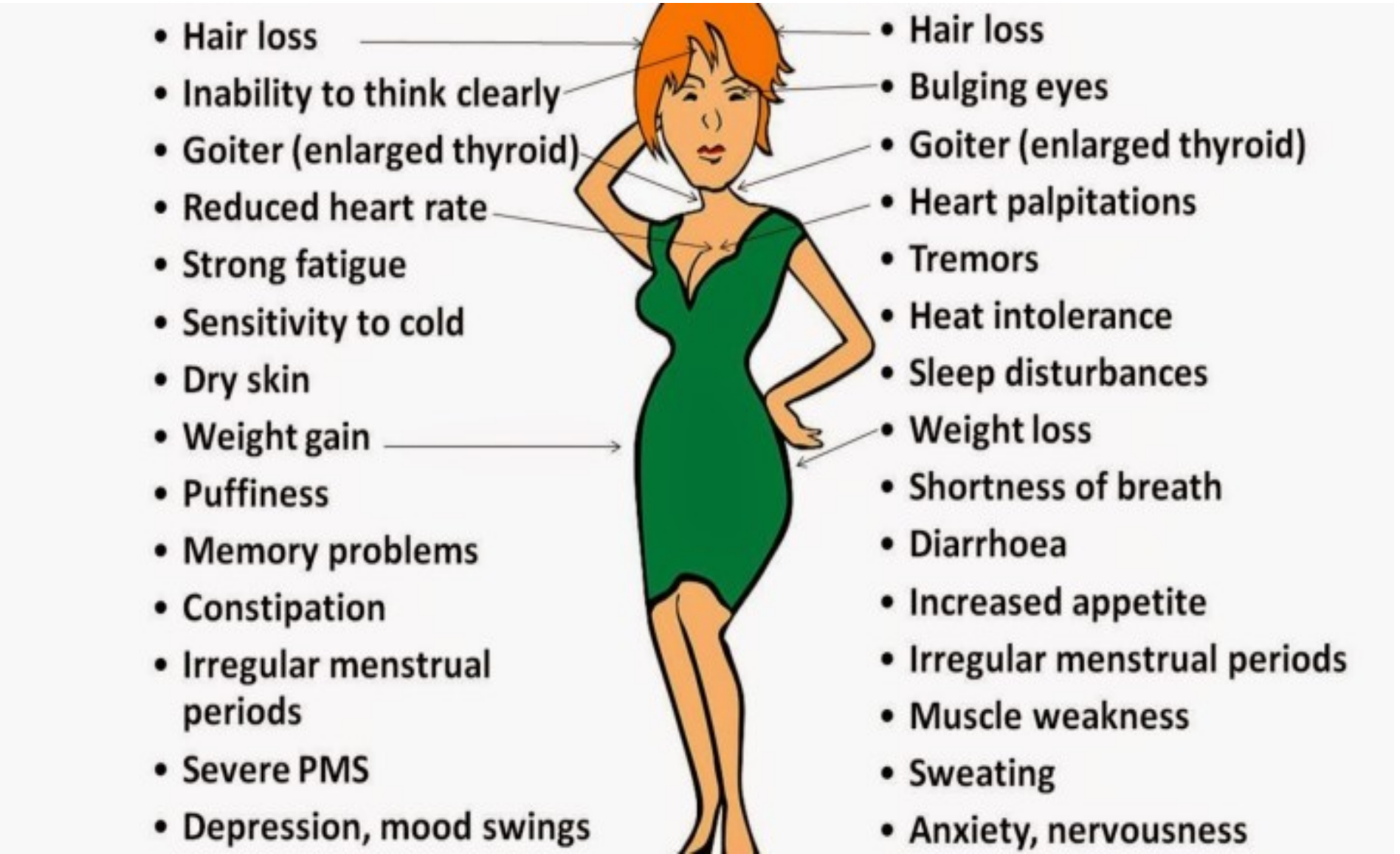
- through hypothalamo-hypophyseal portal circulation.

- TRH is stimulated by cold & inhibited by cortisol.

2- Negative feedback effect of thyroid hormone: more directly at the level of anterior pituitary thyrotrop than on the level of anterior hypothalamus



THYROID DISORDERS





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